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Project Overview

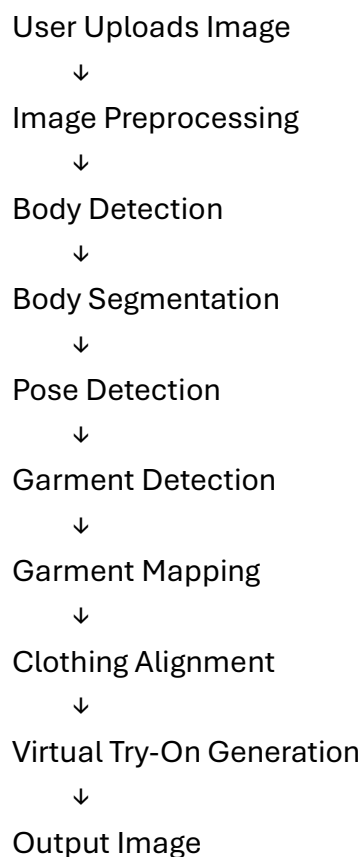
AI-Based Virtual Try-On System(Member 5)

The Virtual Try-On System is an AI-powered application that allows users to visualize how a garment would look on them without physically wearing it. The system uses computer vision and deep learning techniques to analyze the user's image, detect body structure and pose, and digitally fit selected clothing onto the person's image.

The project combines multiple technologies such as body detection, human segmentation, pose estimation, garment mapping, and image generation to create realistic try-on results. By understanding the user's body shape and posture, the system aligns the garment accurately and generates a natural-looking output.

This project aims to improve the online shopping experience by helping users preview clothing before purchase, reducing uncertainty and increasing customer confidence. The workflow involves image preprocessing, body analysis, garment fitting, and final image generation to produce an effective virtual try-on experience.

1. End-to-End Virtual Try-On Workflow



2 Workflow Explanation

User Image Upload

The user provides a full-body image along with the garment image that needs to be virtually worn.

Image Preprocessing

The uploaded image is resized and cleaned to improve processing quality.

Body Detection

The system identifies the person present in the image.

Body Segmentation

Different body regions such as arms, legs, face, and torso are separated for accurate clothing placement.

Pose Detection

Body key points like shoulders, elbows, knees, and hips are detected to understand body posture.

Garment Mapping

The selected garment is mapped according to the body structure and pose information.

Clothing Alignment

The garment is adjusted to match the user's body size and orientation.

Virtual Try-On Generation

The system combines the body image and garment image to create a realistic try-on result.

Final Output

The generated image displays the user wearing the selected garment.

Practical Implementation

To validate the workflow, a pose detection model was implemented using MediaPipe. The model identifies key body landmarks and helps understand human posture, which is an important step before garment mapping in a virtual try-on system.

```
1  import cv2
2  import mediapipe as mp
3  import matplotlib.pyplot as plt
4
5  # Load image
6  image_path = r"C:\Users\Hansika\Downloads\day12.jpeg"
7  image = cv2.imread(image_path)
8
9  if image is None:
10     print("Image not found!")
11     exit()
12
13  # Convert to RGB
14  rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
15
16  # MediaPipe Pose
17  mp_pose = mp.solutions.pose
18  mp_draw = mp.solutions.drawing_utils
19
20  with mp_pose.Pose(static_image_mode=True) as pose:
21     results = pose.process(rgb)
22
23     if results.pose_landmarks:
24         mp_draw.draw_landmarks(
25             image,
26             results.pose_landmarks,
27             mp_pose.POSE_CONNECTIONS
28         )
29
30  # Save output
31  cv2.imwrite("pose_output.jpg", image)
32
33  # Display
34  plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
35  plt.axis("off")
36  plt.title("Pose Detection Output")
37  plt.show()
```

Pose Detection Output



The pose detection model successfully identified major body landmarks such as shoulders, elbows, hips, knees, and ankles. The detected skeleton structure can be used as input for garment alignment and virtual try-on processing.

3 Model Comparison Table

Feature	MediaPipe	OpenPose	Detectron2	SMPL
Main Purpose	Pose Detection	Human Pose Estimation	Segmentation & Detection	3D Human Modeling
Speed	Fast	Moderate	Moderate	Moderate
Accuracy	Good	High	High	Very High
Real-Time Support	Yes	Limited	No	No
Easy to Use	Very Easy	Medium	Medium	Difficult
Body Keypoints	Yes	Yes	Yes	Yes
Segmentation	No	No	Yes	No
3D Modeling	No	No	No	Yes
Best Use Case	Pose Tracking	Human Skeleton Detection	Segmentation Tasks	Avatar Creation

4. Dataset Documentation

Dataset Name	Use Case
DeepFashion	Clothing Recognition and Fashion Analysis
VITON Dataset	Virtual Try-On Training
Dress Code Dataset	Garment Transfer Research
CIHP Dataset	Human Parsing and Segmentation
ATR Dataset	Body Segmentation
COCO Keypoints	Human Pose Detection
MPII Human Pose	Pose Estimation
ModaNet	Fashion Object Detection

Dataset Description

DeepFashion

A large fashion dataset containing clothing images, attributes, and landmarks.

VITON Dataset

Used for training virtual try-on models and garment transfer systems.

Dress Code Dataset

Contains paired clothing and human images for virtual fitting applications.

CIHP Dataset

Provides detailed human parsing labels for body segmentation.

COCO Keypoints

Widely used dataset for human pose estimation tasks.

MPII Human Pose

Contains images with annotated human body joints.

→Challenges Faced

- Understanding different models and their uses took some research.
- Collecting and organizing datasets from various sources was time-consuming.
- Comparing multiple frameworks required careful analysis.

Through this task, I gained a better understanding of how a virtual try-on system works and the technologies involved in it. I studied different frameworks, explored useful datasets, and designed the overall workflow of the system. This helped me understand the role of body detection, segmentation, pose estimation, and garment mapping in creating a virtual try-on application. The research and documentation completed in this task can be used as a foundation for further development and implementation of the project.